Yousef Alsayed Abdelfattah

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PROFESSIONAL SUMMARY

AI Engineer with a B.Sc. in Mechatronics, Robotics, and Automation Engineering. Experienced in building RAG pipelines, fine-tuning LLMs, and multi-agent systems using LangChain, Hugging Face, and LLMS.

EXPERIENCE

AI Software Engineer Nov 2024 – Jan 2025
Outlier AI • Remote

- Evaluated and ranked LLM-generated responses across diverse domains to improve model accuracy and robustness.
- Identified failure modes and collaborated with AI trainers to enhance prompt-response alignment.

PLC Automation Engineer (Internship) Sep 2023
Siemens • Cairo, Egypt

- Designed and implemented industrial automation logic using Siemens TIA Portal for S7-1200 and S7-1500 PLCs with Ladder Logic programming.
- Debugged and troubleshot automation processes, gaining hands-on experience with real-world factory workflows and system integration.

CNC Programming & Operation (Internship) Sep 2022
DMG MORI • Cairo, Egypt

- Developed G-code programs for turning and milling operations; set up and operated CNC machines to precision engineering standards.

PROJECTS

Price Prediction & Autonomous Deal Agent

Built a price prediction pipeline from short product descriptions, fine-tuned an open-source LLM with LoRA/PEFT via Hugging Face to achieve high accuracy at zero cost using Groq, and engineered a multi-agent system that scans listings, detects pricing anomalies, and sends real-time deal alerts.

Medical RAG AI Agent

Architected an end-to-end RAG pipeline using LangChain, ChromaDB, and Hugging Face sentence-transformers for semantic document retrieval, integrated Groq and Ollama (Llama 3.2) as interchangeable LLM backends for cloud and offline inference, added agentic tool-use with web search, PubMed, and OCR capabilities, and delivered a Streamlit web UI with text-to-speech output and configurable parameters.

R2-Waiter

Built a PyQt5 desktop application as my graduation project, integrating an animated robot face (HTML/CSS/JS) that reacts live to ROS velocity commands with eye tracking and facial expressions, combined with a Qt order management interface supporting tray assignment and table selection.

EDUCATION

Bachelor of Science – Mechatronics, Robotics, and Automation Engineering Jun 2024
Misr University for Science and Technology (MUST) • GIZA, Egypt • Sep 2019 – Jun 2024

CERTIFICATIONS

- LLM Engineering: Master AI & Large Language Models – Ed Donner / O'Reilly (2026)
- AI For Everyone – DeepLearning.AI (Coursera) (2025)
- Embedded Systems: AVR – Independent (2023)
- CS50's Introduction to Computer Science(2023)/ CS50's Introduction to Artificial Intelligence(2025)– Harvard / edX (Verified Certificate)
- EF SET English Certificate – C2 Proficient (71/100)

SKILLS

LLMs & AI: RAG Pipelines, AI Agents, Prompt Engineering, Fine-tuning (LoRA/PEFT), Embeddings & Semantic Search
Frameworks: LangChain, LangGraph, FastAPI, Streamlit, Pydantic
Vector Databases: ChromaDB, Sentence-Transformers, Semantic Search, Embedding Models
Languages: Python (primary), C
Tools & DevOps: Git, GitHub, uv, pip, docker

Languages

ARABIC: (Native) **ENGLISH:** (C2)